



THE 4-TIER HETEROGENEOUS SILICON PLACEMENT MAP

Mapping the 7 Agent Organs across Cloud, Linux MPU, Neural Accelerator, and Real-Time MCU Silicon

TIER 1: CLOUD / SERVER

Multi-GPU Cluster · x86 + H100

Stage 4: Deliberation

70B+ Foundation VLM · Task Planning · Open-World Queries

Global Knowledge

Semantic Map Updates · Fleet Experience · Long-Horizon Memory

Silicon Resource Profile:

- **Memory:** Terabytes Host DRAM
- **Clock:** 2.5–3.5 GHz Multi-Socket
- **Latency:** 200–2000 ms (Stochastic)
- **Power:** > 500 Watts (Off-Board)

- **Trust Domain:** **Untrusted** (Network Timeout)

TIER 2: LINUX HOST (MPU)

ARM Cortex-A55/A78 · Linux/POSIX

Stage 2: Ingestion Host

Sensor Fusion · Frame Formatting

Stage 3: Memory / Belief

SE(3) Frame Tree · Temporal Buffer

Stage 7: Governance

Blackbox Ring Buffer · Human Override

Silicon Resource Profile:

- **Memory:** 2–16 GB LPDDR4X/LPDDR5
- **Clock:** 1.2–2.2 GHz Multi-Core
- **Latency:** 10–50 ms (P_{99} Tail Spikes)
- **Power:** 3–15 Watts (On-Board)

- **Trust Domain:** **Untrusted** (Kernel Panics)

TIER 3: NEURAL ENGINE (NPU)

Systolic Array · Tensor Cores

Spatial Encoders (ViT / DINO)

3D Affordance Tokens · Depth Extraction

Stage 5: Planning / Action

Diffusion ACT Policy · $H = 16$ Chunks

Silicon Resource Profile:

- **Memory:** Dedicated SRAM + Shared UMA
- **Throughput:** 8–40 INT8 TOPs
- **Latency:** 5–30 ms (Batch Size = 1)
- **Power:** 2–10 Watts (Thermal Sensitive)

- **Trust Domain:** **Untrusted** (Hallucinations)

TIER 4: SAFETY CORE (MCU)

ARM Cortex-M4/M7 · Bare-Metal / RTOS

Stage 1: Sensing / IMU

Direct ADC/SPI · PTP Timestamps

Stage 6: Reflex Enforcer

1 kHz Control Barrier QP · d_{stop} Envelope

Hardware Watchdog

Direct PWM Gate Cutoff · Zero-Alloc ISR

Silicon Resource Profile:

- **Memory:** 128–512 KB Zero-Wait SRAM
- **Clock:** 100–480 MHz Deterministic
- **Latency:** < 1.0 ms (Jitter < 5 μ s)
- **Power:** 100–400 mW (Static Budget)

- **Trust Domain:** **Trusted** (Inviolable Gatekeeper)

Proposals p_i (Shared SRAM Mailbox / OpenAMP RPMsg)



INTERCONNECT BUS TOPOLOGY & QUALITY-OF-SERVICE (QoS) ARBITRATION:

5G / Wi-Fi 6E Wireless Link

High Jitter ($P_{99} > 450$ ms)

Non-blocking Async Leases

AXI / PCIe Gen3 Crossbar

4–8 GB/s Shared UMA Bus

QoS: Video DMA Throttled

Dedicated SRAM / DMA Mailbox

Zero-Copy Tensor Buffers

Burst Weight Streaming

Shared SRAM / SPI / CAN-FD

Priority 0 (Zero Contention)

< 250 μ s Emergency Reflex